Predictive Analytics – 20 Hrs

Agenda - Day 1	**Introduction** • Machine Learning & Predictive Analysis • What is supervised learning • Data partitioning and holdout samples • Choosing variables (features) • Handling missing data • Visualization and exploration
Agenda - Day 2	**Getting started with Programming language** • Working with Python Programming • Getting Started with IDE (Anaconda- Spyder for Python) • Variables, Data Structure • Control Flow • Modules and Packages
Agenda - Day 3	**Classification and Regression** • Linear Regression • Gradient Descent Algorithm • RMSE, MAPE and R2 value • Implementing Linear Regression with Python • Logistic Regression • Sigmoidal Function • Implementing Logistic Regression with Python • Assessing classification models • Confusion matrix, Recall, Accuracy, F1 Score • ROC Curve Analysis • Misclassification costs • Assessing prediction models • Common metrics • K-Nearest-Neighbors (KNN) • Measuring distance • Choosing k • Generating classifications and predictions
Agenda - Day 4	**Decision Tree** • Decision Tree Introduction • Classification and Regression Trees (CART) • Growing the tree • Avoiding overfit - pruning • Using trees for classifications and predictions

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Agenda - Day 5	**Ensembles** • Combine multiple algorithms • Improve result